

Homework 2: How Many Samples?

BEE 4850/5850, Spring 2027

Due: Thursday 11 February 2027, 9:00pm Eastern

Submit a PDF to Gradescope with **each sub-part tagged**. Untagged submissions lose 10%.

20 points. Problems 1 and 2 are pen-and-paper (14 points). Problem 3 is computational (6 points).

Problem 1 — A Monte Carlo estimate you have to defend (7 points)

You are estimating $m = \mathbb{E}[h(X)]$, the probability that a tide gauge exceeds a threshold in a given year, by drawing $X_1, \dots, X_n$ from a fitted distribution and forming

$$\hat{m}_n = \frac{1}{n} \sum_{i=1}^n h(X_i).$$

(a) (2 points) What does it mean to say $\hat{m}_n$ is an **unbiased** estimator of m ? State what the law of large numbers guarantees about $\hat{m}_n$, and state one thing it does **not** guarantee.

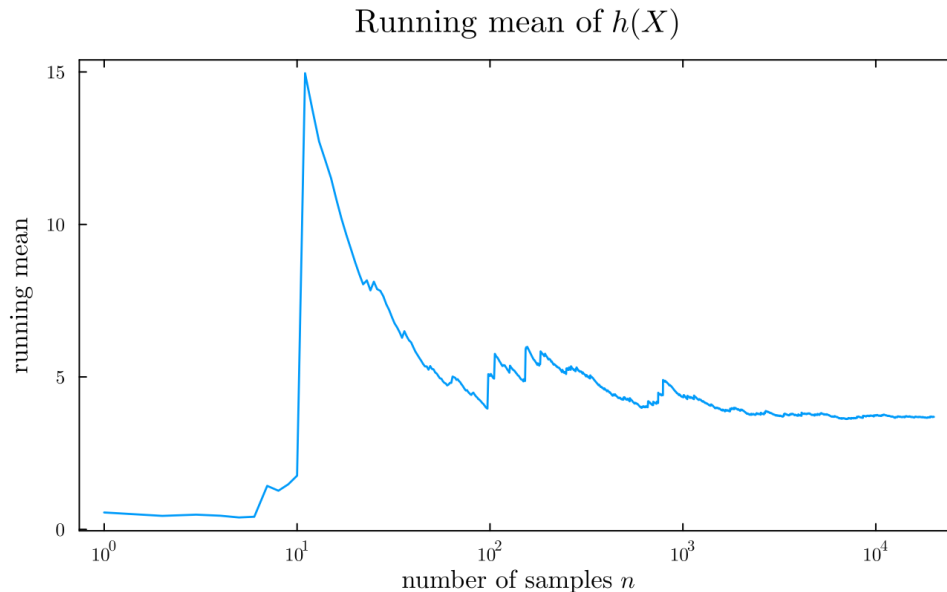
(b) (2 points) You estimate that the standard deviation of $h(X)$ is 5. How many samples do you need for a standard error of 0.1? **Give the relation between n and the standard error**, not only the number.

(c) (3 points) You run the estimator twice, once with $n = 1,000$ and once with $n = 100,000$, and get different answers.

- i. Which do you trust more, and why?
- ii. What would you report alongside the estimate?
- iii. What would make you distrust **both**?

Problem 2 — Short questions (7 points)

(a) (2 points) The figure below shows the running mean $\hat{m}_n$ against n on a log scale. Has the estimator converged? **Justify your answer from features of the plot**, not from the value it appears to settle at.



(b) (2 points) Mark each statement **true** or **false**. For each false statement, correct it in a phrase.

- i. Doubling n halves the Monte Carlo standard error.
- ii. The Monte Carlo standard error tells you how far $\hat{m}_n$ is from m .
- iii. The Monte Carlo standard error can be made arbitrarily small by increasing n .
- iv. If $h(X)$ has infinite variance, the standard error formula still applies.

(c) (3 points) You simulate from a distribution to estimate an expectation. Write down what you are assuming (i) about the sampler and (ii) about h . For **each**, name what goes wrong if the assumption fails.

Problem 3 — Estimating an exceedance probability (6 points)

Use the annual maxima you produced in Homework 1, keeping only years with at least 8,000 recorded hours. (data/annual_maxima.csv is provided if you would rather not recompute them.)

(a) **(2 points)** Fit a normal distribution to the annual maxima by matching the mean and standard deviation. Report both. Then compute, **analytically**, the probability that a year's maximum exceeds **1.5 m** under this fitted distribution.

(b) **(3 points)** Now estimate the same probability by **Monte Carlo**: draw N values from the fitted distribution and take the fraction exceeding 1.5 m. Do this for $N = 10^2, 10^3, 10^4, 10^5$.

For each N , report the estimate **and its standard error**. Present the four results as a small table, and say in one sentence whether the errors behave as Problem 1(b) predicts.

(c) **(1 point)** Compare your answers to the **observed** fraction of years in the record that exceeded 1.5 m. In one sentence: what would it mean if these disagreed badly?